$$
\begin{aligned}
&C_{HD} \to \frac{1}{16\pi^2}\ \Bigg(\frac{2}{27}\sum_p g_1^4\, LF_{3,0}[m_{\tilde d}^p] - \frac{5}{36}\sum_p g_1^4\, LF_{4,-1}[m_{\tilde d}^p] + \\
&\quad \frac{8}{135}\sum_p g_1^4\, LF_{5,-2}[m_{\tilde d}^p] + \frac{2}{9}\sum_p g_1^4\, LF_{3,0}[m_{\tilde e}^p] - \frac{5}{12}\sum_p g_1^4\, LF_{4,-1}[m_{\tilde e}^p] + \\
&\quad \frac{8}{45}\sum_p g_1^4\, LF_{5,-2}[m_{\tilde e}^p] + \Big(-c_{2\gamma}\, g_2^2\, c_\gamma^2\, \overline{y_e}^{pr}\, y_e^{pr} + \frac{1}{36}\sum_p \big(4 g_1^4 + 9 g_2^4 c_{2\gamma}^2\big)\Big)\, LF_{3,0}[m_{\tilde l}^p] + \\
&\quad \Big(c_{2\gamma}\, g_2^2\, c_\gamma^2\, \overline{y_e}^{pr}\, y_e^{pr} - \frac{1}{24}\sum_p \big(5 g_1^4 + 6 g_2^4 c_{2\gamma}^2\big)\Big)\, LF_{4,-1}[m_{\tilde l}^p] + \frac{4}{45}\sum_p g_1^4\, LF_{5,-2}[m_{\tilde l}^p] + \\
&\quad \Big(\frac{1}{27}\sum_p g_1^4 + \frac{3}{4} c_{2\gamma}\, g_2^2\, \big(-4 c_\gamma^2\, \overline{y_d}^{pr}\, y_d^{pr} + 4 s_\gamma^2\, \overline{y_u}^{pr}\, y_u^{pr} + \sum_p c_{2\gamma}\, g_2^2\big)\Big)\, LF_{3,0}[m_{\tilde q}^p] + \\
&\quad \Big(3 c_{2\gamma}\, g_2^2\, \big(c_\gamma^2\, \overline{y_d}^{pr}\, y_d^{pr} - s_\gamma^2\, \overline{y_u}^{pr}\, y_u^{pr}\big) - \frac{1}{72}\sum_p \big(5 g_1^4 + 54 g_2^4 c_{2\gamma}^2\big)\Big)\, LF_{4,-1}[m_{\tilde q}^p] + \\
&\quad \frac{4}{135}\sum_p g_1^4\, LF_{5,-2}[m_{\tilde q}^p] + \frac{8}{27}\sum_p g_1^4\, LF_{3,0}[m_{\tilde u}^p] - \frac{5}{9}\sum_p g_1^4\, LF_{4,-1}[m_{\tilde u}^p] + \frac{32}{135}\sum_p g_1^4\, LF_{5,-2}[m_{\tilde u}^p] + \\
&\quad \frac{1}{576}\Big(g_1^4\big(73 + 9 c_{4\gamma}(-2 + c_{4\gamma}) - 36 s_{2\gamma}^4\big) + 18 g_1^2 g_2^2\big(-3 + c_{4\gamma}(2 + c_{4\gamma}) - 4 s_{2\gamma}^4\big) + \\
&\qquad 9 g_2^4\big((3 + c_{4\gamma})^2 - 4 s_{2\gamma}^4\big)\Big)\, LF_{3,0}[m_\Phi] + \frac{1}{192}\Big(-6 g_1^2 g_2^2\big(-3 + c_{4\gamma}(2 + c_{4\gamma}) - 4 s_{2\gamma}^4\big) + \\
&\qquad 3 g_2^4\big(-(3 + c_{4\gamma})^2 + 4 s_{2\gamma}^4\big) + g_1^4\big(-43 - 3 c_{4\gamma}(-2 + c_{4\gamma}) + 12 s_{2\gamma}^4\big)\Big)\, LF_{4,-1}[m_\Phi] + \\
&\quad \frac{4}{45} g_1^4\, LF_{5,-2}[m_\Phi] + \frac{1}{9} g_1^4\, LF_{3,0}[\tilde\mu] + \frac{1}{6} g_1^4\, LF_{4,-1}[\tilde\mu] - \frac{8}{45} g_1^4\, LF_{5,-2}[\tilde\mu] + \\
&\quad \frac{1}{8} g_1^4\, LF_{2,2,-1}[m_1, \tilde\mu]\, (c_\gamma^2 - s_\gamma^2)^2 + \frac{1}{4} g_1^4\, m_1^2\, LF_{2,2,0}[m_1, \tilde\mu]\, (c_\gamma^2 - s_\gamma^2)^2 + \\
&\quad \frac{17}{8} g_2^4\, LF_{2,2,-1}[m_2, \tilde\mu]\, (c_\gamma^2 - s_\gamma^2)^2 + \frac{1}{4} g_2^4\, m_2^2\, LF_{2,2,0}[m_2, \tilde\mu]\, (c_\gamma^2 - s_\gamma^2)^2 - \\
&\quad 4 g_2^4\, LF_{3,2,-2}[m_2, \tilde\mu]\, (c_\gamma^2 - s_\gamma^2)^2 + 3 g_2^4\, m_2^2\, LF_{3,2,-1}[m_2, \tilde\mu]\, (c_\gamma^2 - s_\gamma^2)^2 + \\
&\quad 2 g_2^4\, LF_{3,3,-3}[m_2, \tilde\mu]\, (c_\gamma^2 - s_\gamma^2)^2 - 2 g_2^4\, m_2^2\, LF_{3,3,-2}[m_2, \tilde\mu]\, (c_\gamma^2 - s_\gamma^2)^2 + \\
&\quad 2 g_2^4\, LF_{4,2,-3}[m_2, \tilde\mu]\, (c_\gamma^2 - s_\gamma^2)^2 - 2 g_2^4\, m_2^2\, LF_{4,2,-2}[m_2, \tilde\mu]\, (c_\gamma^2 - s_\gamma^2)^2 + \\
&\quad \frac{2}{3} g_1^2\, \big(c_\gamma\, \overline{a_d}^{pr} - s_\gamma\tilde\mu\, \overline{y_d}^{pr}\big)\big(c_\gamma\, a_d^{pr} - s_\gamma\tilde\mu\, y_d^{pr}\big)\, LF_{2,2,0}[m_{\tilde d}^r, m_{\tilde q}^p] - \\
&\quad \frac{1}{3} g_1^2\, \big(c_\gamma\, \overline{a_d}^{pr} - s_\gamma\tilde\mu\, \overline{y_d}^{pr}\big)\big(c_\gamma\, a_d^{pr} - s_\gamma\tilde\mu\, y_d^{pr}\big)\, LF_{3,2,-1}[m_{\tilde d}^r, m_{\tilde q}^p] + \\
&\quad \frac{2}{3} g_1^2\, \big(c_\gamma\, \overline{a_e}^{pr} - s_\gamma\tilde\mu\, \overline{y_e}^{pr}\big)\big(c_\gamma\, a_e^{pr} - s_\gamma\tilde\mu\, y_e^{pr}\big)\, LF_{2,2,0}[m_{\tilde e}^r, m_{\tilde l}^p] - \\
&\quad \frac{1}{3} g_1^2\, \big(c_\gamma\, \overline{a_e}^{pr} - s_\gamma\tilde\mu\, \overline{y_e}^{pr}\big)\big(c_\gamma\, a_e^{pr} - s_\gamma\tilde\mu\, y_e^{pr}\big)\, LF_{3,2,-1}[m_{\tilde e}^r, m_{\tilde l}^p] - \\
&\quad \frac{2}{3} g_1^2\, \big(c_\gamma\, \overline{a_e}^{pr} - s_\gamma\tilde\mu\, \overline{y_e}^{pr}\big)\big(c_\gamma\, a_e^{pr} - s_\gamma\tilde\mu\, y_e^{pr}\big)\, LF_{3,1,0}[m_{\tilde l}^p, m_{\tilde e}^r] - \\
&\quad \frac{2}{3} g_1^2\, \big(c_\gamma\, \overline{a_e}^{pr} - s_\gamma\tilde\mu\, \overline{y_e}^{pr}\big)\big(c_\gamma\, a_e^{pr} - s_\gamma\tilde\mu\, y_e^{pr}\big)\, LF_{3,2,-1}[m_{\tilde l}^p, m_{\tilde e}^r] + \\
&\quad \frac{1}{2}\big(3 g_1^2 - c_{2\gamma}\, g_2^2\big)\big(c_\gamma\, \overline{a_e}^{pr} - s_\gamma\tilde\mu\, \overline{y_e}^{pr}\big)\big(c_\gamma\, a_e^{pr} - s_\gamma\tilde\mu\, y_e^{pr}\big)\, LF_{4,1,-1}[m_{\tilde l}^p, m_{\tilde e}^r] - \\
&\quad \frac{2}{3} g_1^2\, \big(c_\gamma\, \overline{a_e}^{pr} - s_\gamma\tilde\mu\, \overline{y_e}^{pr}\big)\big(c_\gamma\, a_e^{pr} - s_\gamma\tilde\mu\, y_e^{pr}\big)\, LF_{5,1,-2}[m_{\tilde l}^p, m_{\tilde e}^r] + \\
&\quad c_\gamma^4\, \overline{y_e}^{pr}\, \overline{y_e}^{st}\, y_e^{pt}\, y_e^{sr}\, LF_{2,1,0}[m_{\tilde l}^p, m_{\tilde l}^s] - c_\gamma^4\, \overline{y_e}^{pr}\, \overline{y_e}^{st}\, y_e^{pt}\, y_e^{sr}\, LF_{3,1,-1}[m_{\tilde l}^p, m_{\tilde l}^s] - \\
&\quad \frac{2}{3} g_1^2\, \big(c_\gamma\, \overline{a_d}^{pr} - s_\gamma\tilde\mu\, \overline{y_d}^{pr}\big)\big(c_\gamma\, a_d^{pr} - s_\gamma\tilde\mu\, y_d^{pr}\big)\, LF_{3,1,0}[m_{\tilde q}^p, m_{\tilde d}^r] - \\
&\quad \frac{2}{3} g_1^2\, \big(c_\gamma\, \overline{a_d}^{pr} - s_\gamma\tilde\mu\, \overline{y_d}^{pr}\big)\big(c_\gamma\, a_d^{pr} - s_\gamma\tilde\mu\, y_d^{pr}\big)\, LF_{3,2,-1}[m_{\tilde q}^p, m_{\tilde d}^r] + \\
&\quad \frac{1}{2}\big(5 g_1^2 - 3 c_{2\gamma}\, g_2^2\big)\big(c_\gamma\, \overline{a_d}^{pr} - s_\gamma\tilde\mu\, \overline{y_d}^{pr}\big)\big(c_\gamma\, a_d^{pr} - s_\gamma\tilde\mu\, y_d^{pr}\big)\, LF_{4,1,-1}[m_{\tilde q}^p, m_{\tilde d}^r] - \\
&\quad 2 g_1^2\, \big(c_\gamma\, \overline{a_d}^{pr} - s_\gamma\tilde\mu\, \overline{y_d}^{pr}\big)\big(c_\gamma\, a_d^{pr} - s_\gamma\tilde\mu\, y_d^{pr}\big)\, LF_{5,1,-2}[m_{\tilde q}^p, m_{\tilde d}^r] + \\
&\quad 3\Big(c_\gamma^2\, \overline{y_d}^{pr}\, y_d^{sr}\big(c_\gamma^2\, \overline{y_d}^{st}\, y_d^{pt} - s_\gamma^2\, \overline{y_u}^{st}\, y_u^{pt}\big) + s_\gamma^4\, \overline{y_u}^{pr}\, \overline{y_u}^{st}\, y_u^{pt}\, y_u^{sr}\Big)\, LF_{2,1,0}[m_{\tilde q}^p, m_{\tilde q}^s] - \\
&\quad 3\Big(c_\gamma^2\, \overline{y_d}^{pr}\, y_d^{sr}\big(c_\gamma^2\, \overline{y_d}^{st}\, y_d^{pt} - s_\gamma^2\, \overline{y_u}^{st}\, y_u^{pt}\big) + s_\gamma^4\, \overline{y_u}^{pr}\, \overline{y_u}^{st}\, y_u^{pt}\, y_u^{sr}\Big)\, LF_{3,1,-1}[m_{\tilde q}^p, m_{\tilde q}^s] - \\
&\quad \frac{1}{3} g_1^2\, \big(s_\gamma\, \overline{a_u}^{pr} - \tilde\mu\, c_\gamma\, \overline{y_u}^{pr}\big)\big(s_\gamma\, a_u^{pr} - \tilde\mu\, c_\gamma\, y_u^{pr}\big)\, LF_{2,2,0}[m_{\tilde q}^p, m_{\tilde u}^r] + \\
&\quad 3 c_{2\gamma}\, g_2^2\, \big(s_\gamma\, \overline{a_u}^{pr} - \tilde\mu\, c_\gamma\, \overline{y_u}^{pr}\big)\big(s_\gamma\, a_u^{pr} - \tilde\mu\, c_\gamma\, y_u^{pr}\big)\, LF_{3,1,0}[m_{\tilde q}^p, m_{\tilde u}^r] + \\
&\quad \frac{1}{6}\big(g_1^2 - 9 c_{2\gamma}\, g_2^2\big)\big(s_\gamma\, \overline{a_u}^{pr} - \tilde\mu\, c_\gamma\, \overline{y_u}^{pr}\big)\big(s_\gamma\, a_u^{pr} - \tilde\mu\, c_\gamma\, y_u^{pr}\big)\, LF_{3,2,-1}[m_{\tilde q}^p, m_{\tilde u}^r] - \\
&\quad 3 c_{2\gamma}\, g_2^2\, \big(s_\gamma\, \overline{a_u}^{pr} - \tilde\mu\, c_\gamma\, \overline{y_u}^{pr}\big)\big(s_\gamma\, a_u^{pr} - \tilde\mu\, c_\gamma\, y_u^{pr}\big)\, LF_{4,1,-1}[m_{\tilde q}^p, m_{\tilde u}^r] - \\
&\quad 3 s_\gamma^2\, c_\gamma^2\, \overline{y_d}^{pr}\, y_d^{sr}\, \overline{y_u}^{st}\, y_u^{pt}\, LF_{2,1,0}[m_{\tilde q}^s, m_{\tilde q}^p] + 3 s_\gamma^2\, c_\gamma^2\, \overline{y_d}^{pr}\, y_d^{sr}\, \overline{y_u}^{st}\, y_u^{pt}\, LF_{3,1,-1}[m_{\tilde q}^s, m_{\tilde q}^p] + \\
&\quad \frac{1}{3} g_1^2\, \big(s_\gamma\, \overline{a_u}^{pr} - \tilde\mu\, c_\gamma\, \overline{y_u}^{pr}\big)\big(s_\gamma\, a_u^{pr} - \tilde\mu\, c_\gamma\, y_u^{pr}\big)\, LF_{3,1,0}[m_{\tilde u}^r, m_{\tilde q}^p] + \\
&\quad \frac{1}{3} g_1^2\, \big(s_\gamma\, \overline{a_u}^{pr} - \tilde\mu\, c_\gamma\, \overline{y_u}^{pr}\big)\big(s_\gamma\, a_u^{pr} - \tilde\mu\, c_\gamma\, y_u^{pr}\big)\, LF_{3,2,-1}[m_{\tilde u}^r, m_{\tilde q}^p] + \\
&\quad g_1^2\, \big(s_\gamma\, \overline{a_u}^{pr} - \tilde\mu\, c_\gamma\, \overline{y_u}^{pr}\big)\big(s_\gamma\, a_u^{pr} - \tilde\mu\, c_\gamma\, y_u^{pr}\big)\, LF_{4,1,-1}[m_{\tilde u}^r, m_{\tilde q}^p] - \\
&\quad 2 g_1^2\, \big(s_\gamma\, \overline{a_u}^{pr} - \tilde\mu\, c_\gamma\, \overline{y_u}^{pr}\big)\big(s_\gamma\, a_u^{pr} - \tilde\mu\, c_\gamma\, y_u^{pr}\big)\, LF_{5,1,-2}[m_{\tilde u}^r, m_{\tilde q}^p] + \\
&\quad \frac{1}{4} g_1^4\, LF_{3,2,-2}[\tilde\mu, m_1]\, (c_\gamma^2 - s_\gamma^2)^2 - \frac{1}{4} g_1^4\, m_1^2\, LF_{3,2,-1}[\tilde\mu, m_1]\, (c_\gamma^2 - s_\gamma^2)^2 - \\
&\quad \frac{5}{6} g_1^4\, LF_{4,1,-2}[\tilde\mu, m_1] - m_1\, s_\gamma\, \tilde\mu\, c_\gamma\, g_1^4\, LF_{4,1,-1}[\tilde\mu, m_1] - \frac{1}{4} g_1^4\, LF_{4,2,-3}[\tilde\mu, m_1]\, (c_\gamma^2 - s_\gamma^2)^2 + \\
&\quad \frac{1}{4} g_1^4\, m_1^2\, LF_{4,2,-2}[\tilde\mu, m_1]\, (c_\gamma^2 - s_\gamma^2)^2 + \frac{2}{3} g_1^4\, LF_{5,1,-3}[\tilde\mu, m_1] + \frac{4}{3} m_1\, s_\gamma\, \tilde\mu\, c_\gamma\, g_1^4\, LF_{5,1,-2}[\tilde\mu, m_1] - \\
&\quad \frac{3}{4} g_2^4\, LF_{3,2,-2}[\tilde\mu, m_2]\, (c_\gamma^2 - s_\gamma^2)^2 - \frac{1}{4} g_2^4\, m_2^2\, LF_{3,2,-1}[\tilde\mu, m_2]\, (c_\gamma^2 - s_\gamma^2)^2 - \\
&\quad \frac{5}{2} g_1^2\, g_2^2\, LF_{4,1,-2}[\tilde\mu, m_2] - 3 m_2\, s_\gamma\, \tilde\mu\, c_\gamma\, g_1^2\, g_2^2\, LF_{4,1,-1}[\tilde\mu, m_2] - \\
&\quad \frac{1}{4} g_2^4\, LF_{4,2,-3}[\tilde\mu, m_2]\, (c_\gamma^2 - s_\gamma^2)^2 + \frac{1}{4} g_2^4\, m_2^2\, LF_{4,2,-2}[\tilde\mu, m_2]\, (c_\gamma^2 - s_\gamma^2)^2 + \\
&\quad 2 g_1^2\, g_2^2\, LF_{5,1,-3}[\tilde\mu, m_2] + 4 m_2\, s_\gamma\, \tilde\mu\, c_\gamma\, g_1^2\, g_2^2\, LF_{5,1,-2}[\tilde\mu, m_2] - 4 g_1^2\, g_2^2\, LF_{2,2,1,-2}[m_2, \tilde\mu, m_1] + \\
&\quad g_1^2\, g_2^2\, \big(-3 m_1 m_2 - 2 s_\gamma\, \tilde\mu\, c_\gamma\, (7 m_1 + 5 m_2) - 12 s_\gamma^2\, \tilde\mu^2\, c_\gamma^2\big)\, LF_{2,2,1,-1}[m_2, \tilde\mu, m_1] - \\
&\quad 8 m_1 m_2\, g_1^2\, g_2^2\, s_\gamma^2\, \tilde\mu^2\, c_\gamma^2\, LF_{2,2,1,0}[m_2, \tilde\mu, m_1] + 2 g_1^2\, g_2^2\, LF_{3,2,1,-3}[m_2, \tilde\mu, m_1] + \\
&\quad 2 g_1^2\, g_2^2\, \big(m_1 m_2 + 4 s_\gamma\, \tilde\mu\, c_\gamma\, (m_1 + m_2) + 4 s_\gamma^2\, \tilde\mu^2\, c_\gamma^2\big)\, LF_{3,2,1,-2}[m_2, \tilde\mu, m_1] + \\
&\quad 8 m_1 m_2\, g_1^2\, g_2^2\, s_\gamma^2\, \tilde\mu^2\, c_\gamma^2\, LF_{3,2,1,-1}[m_2, \tilde\mu, m_1] + \\
&\quad c_\gamma^2\, \overline{y_e}^{st}\, y_e^{pt}\, \big(s_\gamma^2\, \tilde\mu^2\, \overline{y_e}^{pr}\, y_e^{sr} + c_\gamma\, \overline{a_e}^{pr}\, (c_\gamma\, a_e^{sr} - s_\gamma\, \tilde\mu\, y_e^{sr})\big)\, LF_{2,2,1,-1}[m_{\tilde l}^p, m_{\tilde l}^s, m_{\tilde e}^r] - \\
&\quad s_\gamma\, \tilde\mu\, c_\gamma^3\, \overline{y_e}^{pr}\, \overline{y_e}^{st}\, y_e^{sr}\, a_e^{pt}\, LF_{2,2,1,-1}[m_{\tilde l}^p, m_{\tilde l}^s, m_{\tilde e}^t] + \\
&\quad 3\Big(c_\gamma\, \overline{a_d}^{pr}\, \big(c_\gamma^2\, \overline{y_d}^{st}\, y_d^{pt} - s_\gamma^2\, \overline{y_u}^{st}\, y_u^{pt}\big)\big(c_\gamma\, a_d^{sr} - s_\gamma\, \tilde\mu\, y_d^{sr}\big) + \\
&\qquad \tilde\mu\, s_\gamma^2\, \overline{y_d}^{pr}\, \big(\tilde\mu\, c_\gamma^2\, \overline{y_d}^{st}\, y_d^{pt}\, y_d^{sr} + s_\gamma\, \overline{y_u}^{st}\, y_u^{pt}\, (c_\gamma\, a_d^{sr} - s_\gamma\, \tilde\mu\, y_d^{sr})\big)\Big) \\
&\quad\ LF_{2,2,1,-1}[m_{\tilde q}^p, m_{\tilde q}^s, m_{\tilde d}^r] - 3 s_\gamma\, \tilde\mu\, c_\gamma^3\, \overline{y_d}^{pr}\, \overline{y_d}^{st}\, y_d^{sr}\, a_d^{pt}\, LF_{2,2,1,-1}[m_{\tilde q}^p, m_{\tilde q}^s, m_{\tilde d}^t] + \\
&\quad 3 s_\gamma\, \Big(s_\gamma\, \tilde\mu^2\, c_\gamma^2\, \overline{y_u}^{pr}\, \overline{y_u}^{st}\, y_u^{pt}\, y_u^{sr} + \overline{a_u}^{pr}\, \big(c_\gamma^2\, \overline{y_d}^{st}\, y_d^{pt} - s_\gamma^2\, \overline{y_u}^{st}\, y_u^{pt}\big)\big(-s_\gamma\, a_u^{sr} + \tilde\mu\, c_\gamma\, y_u^{sr}\big)\Big) \\
&\quad\ LF_{2,1,1,0}[m_{\tilde q}^p, m_{\tilde q}^s, m_{\tilde u}^r] - 3 s_\gamma\, \Big(s_\gamma\, \tilde\mu^2\, c_\gamma^2\, \overline{y_u}^{pr}\, \overline{y_u}^{st}\, y_u^{pt}\, y_u^{sr} + \\
&\qquad \overline{a_u}^{pr}\, \big(c_\gamma^2\, \overline{y_d}^{st}\, y_d^{pt} - s_\gamma^2\, \overline{y_u}^{st}\, y_u^{pt}\big)\big(-s_\gamma\, a_u^{sr} + \tilde\mu\, c_\gamma\, y_u^{sr}\big)\Big)\, LF_{3,1,1,-1}[m_{\tilde q}^p, m_{\tilde q}^s, m_{\tilde u}^r] + \\
&\quad 3\tilde\mu\, c_\gamma\, \overline{y_u}^{st}\, \big(c_\gamma^2\, \overline{y_d}^{pr}\, y_d^{sr} - s_\gamma^2\, \overline{y_u}^{pr}\, y_u^{sr}\big)\big(s_\gamma\, a_u^{pt} - \tilde\mu\, c_\gamma\, y_u^{pt}\big)\, LF_{2,1,1,0}[m_{\tilde q}^p, m_{\tilde q}^s, m_{\tilde u}^t] + \\
&\quad 3\tilde\mu\, c_\gamma\, \overline{y_u}^{st}\, \big(c_\gamma^2\, \overline{y_d}^{pr}\, y_d^{sr} - s_\gamma^2\, \overline{y_u}^{pr}\, y_u^{sr}\big)\big(-s_\gamma\, a_u^{pt} + \tilde\mu\, c_\gamma\, y_u^{pt}\big)\, LF_{3,1,1,-1}[m_{\tilde q}^p, m_{\tilde q}^s, m_{\tilde u}^t] - \\
&\quad \frac{3}{2} s_\gamma\, \Big(s_\gamma\, \tilde\mu^2\, c_\gamma^2\, \overline{y_u}^{pr}\, \overline{y_u}^{st}\, y_u^{pt}\, y_u^{sr} + \overline{a_u}^{pr}\, \big(c_\gamma^2\, \overline{y_d}^{st}\, y_d^{pt} - s_\gamma^2\, \overline{y_u}^{st}\, y_u^{pt}\big)\big(-s_\gamma\, a_u^{sr} + \tilde\mu\, c_\gamma\, y_u^{sr}\big)\Big) \\
&\quad\ LF_{2,2,1,-1}[m_{\tilde q}^p, m_{\tilde u}^r, m_{\tilde q}^s] + \\
&\quad \frac{3}{2}\tilde\mu\, c_\gamma\, \overline{y_u}^{st}\, \big(c_\gamma^2\, \overline{y_d}^{pr}\, y_d^{sr} - s_\gamma^2\, \overline{y_u}^{pr}\, y_u^{sr}\big)\big(-s_\gamma\, a_u^{pt} + \tilde\mu\, c_\gamma\, y_u^{pt}\big)\, LF_{2,2,1,-1}[m_{\tilde q}^p, m_{\tilde u}^t, m_{\tilde q}^s] + \\
&\quad 3 s_\gamma\, \overline{a_u}^{pr}\, \big(-c_\gamma^2\, \overline{y_d}^{st}\, y_d^{pt} + s_\gamma^2\, \overline{y_u}^{st}\, y_u^{pt}\big)\big(s_\gamma\, a_u^{sr} - \tilde\mu\, c_\gamma\, y_u^{sr}\big)\, LF_{2,1,1,0}[m_{\tilde q}^s, m_{\tilde q}^p, m_{\tilde u}^r] + \\
&\quad 3 s_\gamma\, \overline{a_u}^{pr}\, \big(c_\gamma^2\, \overline{y_d}^{st}\, y_d^{pt} - s_\gamma^2\, \overline{y_u}^{st}\, y_u^{pt}\big)\big(s_\gamma\, a_u^{sr} - \tilde\mu\, c_\gamma\, y_u^{sr}\big)\, LF_{3,1,1,-1}[m_{\tilde q}^s, m_{\tilde q}^p, m_{\tilde u}^r] - \\
&\quad 3\tilde\mu\, c_\gamma\, \overline{y_u}^{st}\, \big(s_\gamma^3\, \overline{y_u}^{pr}\, y_u^{sr}\, a_u^{pt} + c_\gamma^2\, \overline{y_d}^{pr}\, y_d^{sr}\, (-s_\gamma\, a_u^{pt} + \tilde\mu\, c_\gamma\, y_u^{pt})\big)\, LF_{2,1,1,0}[m_{\tilde q}^s, m_{\tilde q}^p, m_{\tilde u}^t] + \\
&\quad 3\tilde\mu\, c_\gamma\, \overline{y_u}^{st}\, \big(s_\gamma^3\, \overline{y_u}^{pr}\, y_u^{sr}\, a_u^{pt} + c_\gamma^2\, \overline{y_d}^{pr}\, y_d^{sr}\, (-s_\gamma\, a_u^{pt} + \tilde\mu\, c_\gamma\, y_u^{pt})\big)\, LF_{3,1,1,-1}[m_{\tilde q}^s, m_{\tilde q}^p, m_{\tilde u}^t] + \\
&\quad \frac{3}{2} s_\gamma\, \overline{a_u}^{pr}\, \big(c_\gamma^2\, \overline{y_d}^{st}\, y_d^{pt} - s_\gamma^2\, \overline{y_u}^{st}\, y_u^{pt}\big)\big(s_\gamma\, a_u^{sr} - \tilde\mu\, c_\gamma\, y_u^{sr}\big)\, LF_{2,2,1,-1}[m_{\tilde q}^s, m_{\tilde u}^r, m_{\tilde q}^p] + \\
&\quad \frac{3}{2}\tilde\mu\, c_\gamma\, \overline{y_u}^{st}\, \big(s_\gamma^3\, \overline{y_u}^{pr}\, y_u^{sr}\, a_u^{pt} + c_\gamma^2\, \overline{y_d}^{pr}\, y_d^{sr}\, (-s_\gamma\, a_u^{pt} + \tilde\mu\, c_\gamma\, y_u^{pt})\big)\, LF_{2,2,1,-1}[m_{\tilde q}^s, m_{\tilde u}^t, m_{\tilde q}^p] + \\
&\quad \frac{7}{4} g_1^2\, g_2^2\, LF_{2,1,1,-1}[\tilde\mu, m_1, m_2] + \frac{1}{2} m_1\, g_1^2\, g_2^2\, (m_2 + 8 s_\gamma\, \tilde\mu\, c_\gamma)\, LF_{2,1,1,0}[\tilde\mu, m_1, m_2] - \\
&\quad \frac{3}{2} g_1^2\, g_2^2\, LF_{3,1,1,-2}[\tilde\mu, m_1, m_2] + \frac{1}{2} g_1^2\, g_2^2\, \big(-m_1 m_2 - 2 s_\gamma\, \tilde\mu\, c_\gamma\, (5 m_1 + m_2) - 8 s_\gamma^2\, \tilde\mu^2\, c_\gamma^2\big) \\
&\quad\ LF_{3,1,1,-1}[\tilde\mu, m_1, m_2] + \frac{1}{2} g_1^2\, g_2^2\, LF_{4,1,1,-3}[\tilde\mu, m_1, m_2] + \\
&\quad \frac{1}{2} g_1^2\, g_2^2\, \big(m_1 m_2 + 4 s_\gamma\, \tilde\mu\, c_\gamma\, (m_1 + m_2) + 4 s_\gamma^2\, \tilde\mu^2\, c_\gamma^2\big)\, LF_{4,1,1,-2}[\tilde\mu, m_1, m_2] + \\
&\quad 2 m_1 m_2\, g_1^2\, g_2^2\, s_\gamma^2\, \tilde\mu^2\, c_\gamma^2\, LF_{4,1,1,-1}[\tilde\mu, m_1, m_2] + 2 g_1^2\, g_2^2\, LF_{3,2,1,-3}[\tilde\mu, m_2, m_1] + \\
&\quad 2 g_1^2\, g_2^2\, \big(m_1 m_2 + 4 s_\gamma\, \tilde\mu\, c_\gamma\, (m_1 + m_2) + 4 s_\gamma^2\, \tilde\mu^2\, c_\gamma^2\big)\, LF_{3,2,1,-2}[\tilde\mu, m_2, m_1] + \\
&\quad 8 m_1 m_2\, g_1^2\, g_2^2\, s_\gamma^2\, \tilde\mu^2\, c_\gamma^2\, LF_{3,2,1,-1}[\tilde\mu, m_2, m_1] + \\
&\quad \frac{1}{2}\Big(s_\gamma^2\, \tilde\mu^2\, \overline{y_e}^{pr}\, \overline{y_e}^{st}\, \big(s_\gamma^2\, \tilde\mu^2\, y_e^{pt}\, y_e^{sr} + c_\gamma\, a_e^{pt}\, (c_\gamma\, a_e^{sr} - 2 s_\gamma\, \tilde\mu\, y_e^{sr})\big) + \\
&\qquad c_\gamma\, \overline{a_e}^{pr}\, \Big(-2 s_\gamma\, \tilde\mu\, \overline{y_e}^{st}\, \big(-c_\gamma\, a_e^{pt} + s_\gamma\, \tilde\mu\, y_e^{pt}\big)\big(-c_\gamma\, a_e^{sr} + s_\gamma\, \tilde\mu\, y_e^{sr}\big) + c_\gamma\, \overline{a_e}^{st} \\
&\qquad\qquad \big(s_\gamma^2\, \tilde\mu^2\, y_e^{pt}\, y_e^{sr} + c_\gamma\, a_e^{pt}\, (c_\gamma\, a_e^{sr} - 2 s_\gamma\, \tilde\mu\, y_e^{sr})\big)\Big)\Big)\, LF_{2,2,1,1,-1}[m_{\tilde l}^p, m_{\tilde l}^s, m_{\tilde e}^r, m_{\tilde e}^t] + \\
&\quad \frac{3}{2}\Big(s_\gamma^2\, \tilde\mu^2\, \overline{y_d}^{pr}\, \overline{y_d}^{st}\, \big(s_\gamma^2\, \tilde\mu^2\, y_d^{pt}\, y_d^{sr} + c_\gamma\, a_d^{pt}\, (c_\gamma\, a_d^{sr} - 2 s_\gamma\, \tilde\mu\, y_d^{sr})\big) + \\
&\qquad c_\gamma\, \overline{a_d}^{pr}\, \Big(-2 s_\gamma\, \tilde\mu\, \overline{y_d}^{st}\, \big(-c_\gamma\, a_d^{pt} + s_\gamma\, \tilde\mu\, y_d^{pt}\big)\big(-c_\gamma\, a_d^{sr} + s_\gamma\, \tilde\mu\, y_d^{sr}\big) + \\
&\qquad\quad c_\gamma\, \overline{a_d}^{st}\, \big(s_\gamma^2\, \tilde\mu^2\, y_d^{pt}\, y_d^{sr} + c_\gamma\, a_d^{pt}\, (c_\gamma\, a_d^{sr} - 2 s_\gamma\, \tilde\mu\, y_d^{sr})\big)\Big)\Big) \\
&\quad\ LF_{2,2,1,1,-1}[m_{\tilde q}^p, m_{\tilde q}^s, m_{\tilde d}^r, m_{\tilde d}^t] - 3 c_\gamma\, \big(c_\gamma\, a_d^{sr} - s_\gamma\, \tilde\mu\, y_d^{sr}\big)\big(s_\gamma\, a_u^{pt} - \tilde\mu\, c_\gamma\, y_u^{pt}\big) \\
&\quad\ \big(s_\gamma\, \tilde\mu^2\, \overline{y_d}^{pr}\, \overline{y_u}^{st} + \overline{a_d}^{pr}\, (s_\gamma\, \overline{a_u}^{st} - \tilde\mu\, c_\gamma\, \overline{y_u}^{st})\big)\, LF_{2,2,1,1,-1}[m_{\tilde q}^p, m_{\tilde q}^s, m_{\tilde d}^r, m_{\tilde u}^t] + \\
&\quad 3\tilde\mu\, s_\gamma^2\, \overline{y_d}^{st}\, \overline{a_u}^{pr}\, \big(c_\gamma\, a_d^{pt} - s_\gamma\, \tilde\mu\, y_d^{pt}\big)\big(s_\gamma\, a_u^{sr} - \tilde\mu\, c_\gamma\, y_u^{sr}\big)\, LF_{2,2,1,1,-1}[m_{\tilde q}^p, m_{\tilde q}^s, m_{\tilde d}^t, m_{\tilde u}^r] + \\
&\quad 3\Big(\tilde\mu^2\, c_\gamma^2\, \overline{y_u}^{pr}\, \overline{y_u}^{st}\, \big(\tilde\mu^2\, c_\gamma^2\, y_u^{pt}\, y_u^{sr} + s_\gamma\, a_u^{pt}\, (s_\gamma\, a_u^{sr} - \tilde\mu\, c_\gamma\, y_u^{sr})\big) + \\
&\qquad s_\gamma\, \overline{a_u}^{pr}\, \Big(-\tilde\mu\, c_\gamma\, \overline{y_u}^{st}\, \big(-s_\gamma\, a_u^{pt} + \tilde\mu\, c_\gamma\, y_u^{pt}\big)\big(-s_\gamma\, a_u^{sr} + \tilde\mu\, c_\gamma\, y_u^{sr}\big) + s_\gamma\, \overline{a_u}^{st} \\
&\qquad\qquad \big(\tilde\mu^2\, c_\gamma^2\, y_u^{pt}\, y_u^{sr} + s_\gamma\, a_u^{pt}\, (s_\gamma\, a_u^{sr} - \tilde\mu\, c_\gamma\, y_u^{sr})\big)\Big)\Big)\, LF_{2,1,1,1,0}[m_{\tilde q}^p, m_{\tilde q}^s, m_{\tilde u}^r, m_{\tilde u}^t] - \\
&\quad 3\Big(\tilde\mu^2\, c_\gamma^2\, \overline{y_u}^{pr}\, \overline{y_u}^{st}\, \big(\tilde\mu^2\, c_\gamma^2\, y_u^{pt}\, y_u^{sr} + s_\gamma\, a_u^{pt}\, (s_\gamma\, a_u^{sr} - \tilde\mu\, c_\gamma\, y_u^{sr})\big) + \\
&\qquad s_\gamma\, \overline{a_u}^{pr}\, \Big(-\tilde\mu\, c_\gamma\, \overline{y_u}^{st}\, \big(-s_\gamma\, a_u^{pt} + \tilde\mu\, c_\gamma\, y_u^{pt}\big)\big(-s_\gamma\, a_u^{sr} + \tilde\mu\, c_\gamma\, y_u^{sr}\big) + s_\gamma\, \overline{a_u}^{st} \\
&\qquad\qquad \big(\tilde\mu^2\, c_\gamma^2\, y_u^{pt}\, y_u^{sr} + s_\gamma\, a_u^{pt}\, (s_\gamma\, a_u^{sr} - \tilde\mu\, c_\gamma\, y_u^{sr})\big)\Big)\Big)\, LF_{3,1,1,1,-1}[m_{\tilde q}^p, m_{\tilde q}^s, m_{\tilde u}^r, m_{\tilde u}^t] - \\
&\quad \frac{3}{2}\Big(\tilde\mu^2\, c_\gamma^2\, \overline{y_u}^{pr}\, \overline{y_u}^{st}\, \big(\tilde\mu^2\, c_\gamma^2\, y_u^{pt}\, y_u^{sr} + s_\gamma\, a_u^{pt}\, (s_\gamma\, a_u^{sr} - \tilde\mu\, c_\gamma\, y_u^{sr})\big) + \\
&\qquad s_\gamma\, \overline{a_u}^{pr}\, \Big(-\tilde\mu\, c_\gamma\, \overline{y_u}^{st}\, \big(-s_\gamma\, a_u^{pt} + \tilde\mu\, c_\gamma\, y_u^{pt}\big)\big(-s_\gamma\, a_u^{sr} + \tilde\mu\, c_\gamma\, y_u^{sr}\big) + s_\gamma\, \overline{a_u}^{st} \\
&\qquad\qquad \big(\tilde\mu^2\, c_\gamma^2\, y_u^{pt}\, y_u^{sr} + s_\gamma\, a_u^{pt}\, (s_\gamma\, a_u^{sr} - \tilde\mu\, c_\gamma\, y_u^{sr})\big)\Big)\Big)\, LF_{2,2,1,1,-1}[m_{\tilde q}^p, m_{\tilde u}^r, m_{\tilde q}^s, m_{\tilde u}^t] - \\
&\quad \frac{3}{2}\Big(\tilde\mu^2\, c_\gamma^2\, \overline{y_u}^{pr}\, \overline{y_u}^{st}\, \big(\tilde\mu^2\, c_\gamma^2\, y_u^{pt}\, y_u^{sr} + s_\gamma\, a_u^{pt}\, (s_\gamma\, a_u^{sr} - \tilde\mu\, c_\gamma\, y_u^{sr})\big) + \\
&\qquad s_\gamma\, \overline{a_u}^{pr}\, \Big(-\tilde\mu\, c_\gamma\, \overline{y_u}^{st}\, \big(-s_\gamma\, a_u^{pt} + \tilde\mu\, c_\gamma\, y_u^{pt}\big)\big(-s_\gamma\, a_u^{sr} + \tilde\mu\, c_\gamma\, y_u^{sr}\big) + \\
&\qquad\quad s_\gamma\, \overline{a_u}^{st}\, \big(\tilde\mu^2\, c_\gamma^2\, y_u^{pt}\, y_u^{sr} + s_\gamma\, a_u^{pt}\, (s_\gamma\, a_u^{sr} - \tilde\mu\, c_\gamma\, y_u^{sr})\big)\Big)\Big) \\
&\quad\ LF_{2,2,1,1,-1}[m_{\tilde q}^p, m_{\tilde u}^t, m_{\tilde q}^s, m_{\tilde u}^r] - 3 s_\gamma\, \tilde\mu\, c_\gamma\, \Big(\tilde\mu^2\, c_\gamma^2\, \overline{y_u}^{pr}\, \overline{y_u}^{st}\, y_u^{sr}\, a_u^{pt} + \\
&\qquad \overline{a_u}^{pr}\, \big(s_\gamma^2\, y_u^{sr}\, \overline{a_u}^{st}\, a_u^{pt} + \overline{y_u}^{st}\, (s_\gamma\, a_u^{pt} - \tilde\mu\, c_\gamma\, y_u^{pt})(s_\gamma\, a_u^{sr} - \tilde\mu\, c_\gamma\, y_u^{sr})\big)\Big) \\
&\quad\ LF_{2,1,1,1,0}[m_{\tilde q}^s, m_{\tilde q}^p, m_{\tilde u}^r, m_{\tilde u}^t] + 3 s_\gamma\, \tilde\mu\, c_\gamma\, \Big(\tilde\mu^2\, c_\gamma^2\, \overline{y_u}^{pr}\, \overline{y_u}^{st}\, y_u^{sr}\, a_u^{pt} + \\
&\qquad \overline{a_u}^{pr}\, \big(s_\gamma^2\, y_u^{sr}\, \overline{a_u}^{st}\, a_u^{pt} + \overline{y_u}^{st}\, (s_\gamma\, a_u^{pt} - \tilde\mu\, c_\gamma\, y_u^{pt})(s_\gamma\, a_u^{sr} - \tilde\mu\, c_\gamma\, y_u^{sr})\big)\Big) \\
&\quad\ LF_{3,1,1,1,-1}[m_{\tilde q}^s, m_{\tilde q}^p, m_{\tilde u}^r, m_{\tilde u}^t] + \frac{3}{2} s_\gamma\, \tilde\mu\, c_\gamma\, \Big(\tilde\mu^2\, c_\gamma^2\, \overline{y_u}^{pr}\, \overline{y_u}^{st}\, y_u^{sr}\, a_u^{pt} + \\
&\qquad \overline{a_u}^{pr}\, \big(s_\gamma^2\, y_u^{sr}\, \overline{a_u}^{st}\, a_u^{pt} + \overline{y_u}^{st}\, (s_\gamma\, a_u^{pt} - \tilde\mu\, c_\gamma\, y_u^{pt})(s_\gamma\, a_u^{sr} - \tilde\mu\, c_\gamma\, y_u^{sr})\big)\Big) \\
&\quad\ LF_{2,2,1,1,-1}[m_{\tilde q}^s, m_{\tilde u}^r, m_{\tilde q}^p, m_{\tilde u}^t] + \frac{3}{2} s_\gamma\, \tilde\mu\, c_\gamma\, \Big(\tilde\mu^2\, c_\gamma^2\, \overline{y_u}^{pr}\, \overline{y_u}^{st}\, y_u^{sr}\, a_u^{pt} + \overline{a_u}^{pr}\, \big(s_\gamma^2\, y_u^{sr}\, \overline{a_u}^{st}\, a_u^{pt} + \\
&\qquad\quad \overline{y_u}^{st}\, (s_\gamma\, a_u^{pt} - \tilde\mu\, c_\gamma\, y_u^{pt})(s_\gamma\, a_u^{sr} - \tilde\mu\, c_\gamma\, y_u^{sr})\big)\Big)\, LF_{2,2,1,1,-1}[m_{\tilde q}^s, m_{\tilde u}^t, m_{\tilde q}^p, m_{\tilde u}^r]\Bigg)
\end{aligned}
$$